Concept Question 8-4: Can total internal reflection take place for a wave incident from medium 1 (with n_1) onto medium 2 (with n_2) when $n_2 > n_1$?

Concept Question 8-5: What is the difference between the boundary conditions applied in Section 8-1.1 for normal incidence and those applied in Section 8-4.1 for oblique incidence with perpendicular polarization?

Concept Question 8-6: Why is the Brewster angle also called the polarizing angle?

Concept Question 8-7: At the boundary, the vector sum of the tangential components of the incident and reflected electric fields has to equal the tangential component of the transmitted electric field. For $\varepsilon_{r_1} = 1$ and $\varepsilon_{r_2} = 16$, determine the Brewster angle and then verify the validity of the preceding statement by sketching to scale the tangential components of the three electric fields at the Brewster angle.

Exercise 8-7: A wave in air is incident upon a soil surface at $\theta_i = 50°$. If soil has $\varepsilon_r = 4$ and $\mu_r = 1$, determine $\Gamma_\perp$, $\tau_\perp$, $\Gamma_\parallel$, and $\tau_\parallel$.

Answer: $\Gamma_\perp = -0.48$, $\tau_\perp = 0.52$, $\Gamma_\parallel = -0.16$, $\tau_\parallel = 0.58$. (See EM.)

Exercise 8-8: Determine the Brewster angle for the boundary of Exercise 8.7.

Answer: $\theta_B = 63.4°$. (See EM.)

Exercise 8-9: Show that the incident, reflected, and transmitted electric and magnetic fields given by Eqs. (8.65a) through (8.65f) all have the same exponential phase function along the x direction.

Answer: With the help of Eqs. (8.55) and (8.56), all six fields are shown to vary as $e^{-jk_1 x \sin\theta_i}$. (See EM.)

8-5 Reflectivity and Transmissivity

The reflection and transmission coefficients derived earlier are ratios of the reflected and transmitted electric field amplitudes to the amplitude of the incident electric field. We now examine power ratios, starting with the perpendicular polarization case. **Figure 8-18** shows a circular beam of electromagnetic

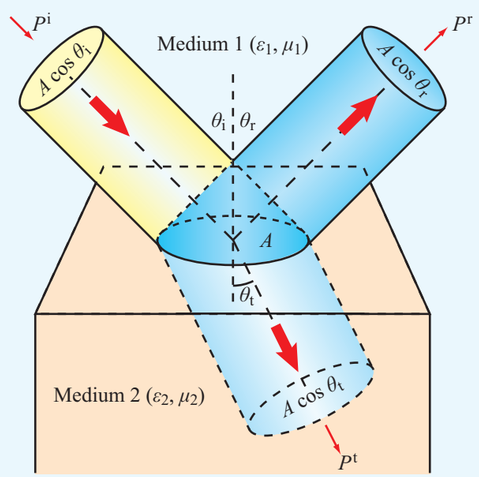


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

energy incident upon the boundary between two contiguous, lossless media. The area of the spot illuminated by the beam is A, and the incident, reflected, and transmitted beams have electric-field amplitudes $E^i_{\perp 0}$, $E^r_{\perp 0}$, and $E^t_{\perp 0}$, respectively. The average power densities carried by the incident, reflected, and transmitted beams are

$$S^i_\perp = \frac{|E^i_{\perp 0}|^2}{2\eta_1}, \tag{8.73a}$$

$$S^r_\perp = \frac{|E^r_{\perp 0}|^2}{2\eta_1}, \tag{8.73b}$$

$$S^t_\perp = \frac{|E^t_{\perp 0}|^2}{2\eta_2}, \tag{8.73c}$$

where η_1 and η_2 are the intrinsic impedances of media 1 and 2, respectively. The cross-sectional areas of the incident, reflected, and transmitted beams are

$$A_i = A\cos\theta_i, \tag{8.74a}$$

$$A_r = A\cos\theta_r, \tag{8.74b}$$

$$A_t = A\cos\theta_t, \tag{8.74c}$$

概念题 8-4： 波由介质1（折射率 n_1）入射到介质2（折射率 n_2），当 $n_2 > n_1$ 时，能否发生全内反射？

概念题 8-5： 第 8-1.1 节垂直入射所用的边界条件，与第 8-4.1 节垂直极化波斜入射所用的边界条件有何区别？

概念题 8-6： 为什么布儒斯特角又称为起偏角？

概念题 8-7： 在界面处，入射电场与反射电场的切向分量之矢量和，必须等于透射电场的切向分量。已知 $\varepsilon_{r1} = 1$、$\varepsilon_{r2} = 16$，求布儒斯特角，并按比例画出该入射角下三个电场的切向分量，以验证上述关系。

练习 8-7： 空气中的波以 $\theta_i = 50°$ 入射到土壤表面。若土壤 $\varepsilon_r = 4$、$\mu_r = 1$，求 $\Gamma\perp$、$\tau\perp$、$\Gamma\parallel$、$\tau\parallel$。

答案：$\Gamma\perp = -0.48$，$\tau\perp = 0.52$，$\Gamma\parallel = -0.16$，$\tau\parallel = 0.58$。（参见配套 EM 资源。）

练习 8-8： 求练习 8-7 中界面的布儒斯特角。

答案：$\theta_B = 63.4°$。（参见配套 EM 资源。）

练习 8-9： 证明式（8.65a）至（8.65f）中的入射、反射和透射电场及磁场，沿 x 方向具有相同的指数相位因子。

答案：利用式（8.55）、（8.56），可得六个场均含因子 $e^{-jk_1 x \sin\theta_i}$。（参见配套 EM 资源。）

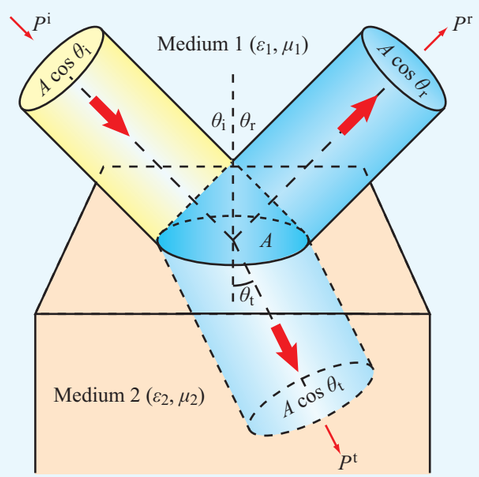


图 8-18 圆形入射波束的反射与透射；它在界面上的照射区域面积为 A。

图例：Medium 1、Medium 2 分别为介质1、介质2。

8-5 反射率与透射率

前面得到的反射系数与透射系数，分别为反射、透射电场振幅与入射电场振幅之比。下面从垂直极化的情况入手，研究功率之比。图 8-18 表示一束圆形截面的电磁波束

入射到两个相邻无损介质的界面。波束照射区域的面积为 A；入射、反射和透射波束的电场振幅分别为 $E^i_{\perp 0}$、$E^r_{\perp 0}$、$E^t_{\perp 0}$。三个波束携带的平均功率密度分别为

$$S^i_\perp = \frac{|E^i_{\perp 0}|^2}{2\eta_1}, \tag{8.73a}$$

$$S^r_\perp = \frac{|E^r_{\perp 0}|^2}{2\eta_1}, \tag{8.73b}$$

$$S^t_\perp = \frac{|E^t_{\perp 0}|^2}{2\eta_2}, \tag{8.73c}$$

其中，η_1 和 η_2 分别为介质1、介质2的本征阻抗。入射、反射和透射波束的横截面积分别为

$$A_i = A\cos\theta_i, \tag{8.74a}$$

$$A_r = A\cos\theta_r, \tag{8.74b}$$

$$A_t = A\cos\theta_t, \tag{8.74c}$$

and the corresponding average powers carried by the beams are

$$P_\perp^{\mathrm{i}} = S_\perp^{\mathrm{i}} A_{\mathrm{i}} = \frac{|E_{\perp 0}^{\mathrm{i}}|^2}{2\eta_1} A\cos\theta_{\mathrm{i}}, \tag{8.75a}$$

$$P_\perp^{\mathrm{r}} = S_\perp^{\mathrm{r}} A_{\mathrm{r}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2}{2\eta_1} A\cos\theta_{\mathrm{r}}, \tag{8.75b}$$

$$P_\perp^{\mathrm{t}} = S_\perp^{\mathrm{t}} A_{\mathrm{t}} = \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{2\eta_2} A\cos\theta_{\mathrm{t}}. \tag{8.75c}$$

The ***reflectivity*** R (also called ***reflectance*** in optics) is defined as the ratio of the reflected to the incident power. The reflectivity for perpendicular polarization is then

$$R_\perp = \frac{P_\perp^{\mathrm{r}}}{P_\perp^{\mathrm{i}}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2\cos\theta_{\mathrm{r}}}{|E_{\perp 0}^{\mathrm{i}}|^2\cos\theta_{\mathrm{i}}} = \left|\frac{E_{\perp 0}^{\mathrm{r}}}{E_{\perp 0}^{\mathrm{i}}}\right|^2, \tag{8.76}$$

where we used the fact that $\theta_{\mathrm{r}} = \theta_{\mathrm{i}}$ in accordance with Snell's law of reflection. The ratio of the reflected to incident electric field amplitudes, $|E_{\perp 0}^{\mathrm{r}}/E_{\perp 0}^{\mathrm{i}}|$, is equal to the magnitude of the reflection coefficient $\Gamma_\perp$. Hence,

$$R_\perp = |\Gamma_\perp|^2, \tag{8.77}$$

and similarly, for parallel polarization

$$R_\parallel = \frac{P_\parallel^{\mathrm{r}}}{P_\parallel^{\mathrm{i}}} = |\Gamma_\parallel|^2. \tag{8.78}$$

The ***transmissivity*** T (or ***transmittance*** in optics) is defined as the ratio of the transmitted power to incident power:

$$T_\perp = \frac{P_\perp^{\mathrm{t}}}{P_\perp^{\mathrm{i}}} = \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{|E_{\perp 0}^{\mathrm{i}}|^2}\frac{\eta_1}{\eta_2}\frac{A\cos\theta_{\mathrm{t}}}{A\cos\theta_{\mathrm{i}}} = |\tau_\perp|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right), \tag{8.79a}$$

$$T_\parallel = \frac{P_\parallel^{\mathrm{t}}}{P_\parallel^{\mathrm{i}}} = |\tau_\parallel|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right). \tag{8.79b}$$

► The incident, reflected, and transmitted waves do not have to obey any such laws as conservation of electric field, conservation of magnetic field, or conservation of power density, but they do have to obey the law of conservation of power. ◄

In fact, in many cases, the transmitted electric field is larger than the incident electric field. Conservation of power requires that the incident power equals the sum of the reflected and transmitted powers. That is, for perpendicular polarization,

$$P_\perp^{\mathrm{i}} = P_\perp^{\mathrm{r}} + P_\perp^{\mathrm{t}}, \tag{8.80}$$

or

$$\frac{|E_{\perp 0}^{\mathrm{i}}|^2}{2\eta_1} A\cos\theta_{\mathrm{i}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2}{2\eta_1} A\cos\theta_{\mathrm{r}} + \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{2\eta_2} A\cos\theta_{\mathrm{t}}. \tag{8.81}$$

Use of Eqs. (8.76), (8.79a), and (8.79b) leads to

$$R_\perp + T_\perp = 1, \tag{8.82a}$$

$$R_\parallel + T_\parallel = 1, \tag{8.82b}$$

or

$$|\Gamma_\perp|^2 + |\tau_\perp|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right) = 1, \tag{8.83a}$$

$$|\Gamma_\parallel|^2 + |\tau_\parallel|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right) = 1. \tag{8.83b}$$

Figure 8-19 shows plots for $(R_\parallel, T_\parallel)$ as a function of θ_{i} for an air–glass interface. Note that the sum of $R_\parallel$ and $T_\parallel$ is always equal to 1, as mandated by Eq. (8.82b). We also note that, at the Brewster angle θ_{B}, $R_\parallel = 0$ and $T_\parallel = 1$.

Table 8-2 provides a summary of the general expressions for Γ, τ, R, and T for both normal and oblique incidence.

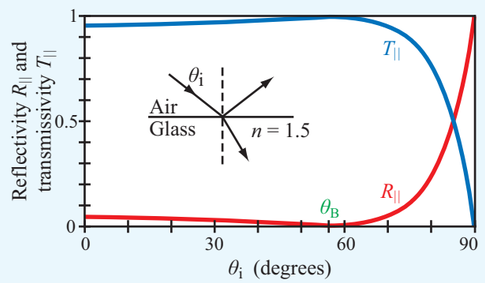


Figure 8-19 Angular plots for $(R_\parallel, T_\parallel)$ for an air–glass interface.

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光束所携带的相应平均功率为

$$P_\perp^{\mathrm{i}} = S_\perp^{\mathrm{i}} A_{\mathrm{i}} = \frac{|E_{\perp 0}^{\mathrm{i}}|^2}{2\eta_1} A\cos\theta_{\mathrm{i}}, \tag{8.75a}$$

$$P_\perp^{\mathrm{r}} = S_\perp^{\mathrm{r}} A_{\mathrm{r}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2}{2\eta_1} A\cos\theta_{\mathrm{r}}, \tag{8.75b}$$

$$P_\perp^{\mathrm{t}} = S_\perp^{\mathrm{t}} A_{\mathrm{t}} = \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{2\eta_2} A\cos\theta_{\mathrm{t}}. \tag{8.75c}$$

反射率 R（在光学中也称为反射率）定义为反射功率与入射功率之比。垂直偏振的反射率为

$$R_\perp = \frac{P_\perp^{\mathrm{r}}}{P_\perp^{\mathrm{i}}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2\cos\theta_{\mathrm{r}}}{|E_{\perp 0}^{\mathrm{i}}|^2\cos\theta_{\mathrm{i}}} = \left|\frac{E_{\perp 0}^{\mathrm{r}}}{E_{\perp 0}^{\mathrm{i}}}\right|^2, \tag{8.76}$$

我们根据斯涅尔反射定律使用了 $\theta_{\mathrm{r}} = \theta_{\mathrm{i}}$ 这一事实。反射电场幅度与入射电场幅度之比 $|E_{\perp 0}^{\mathrm{r}}/E_{\perp 0}^{\mathrm{i}}|$ 等于反射系数 $\Gamma_\perp$ 的大小。因此，

$$R_\perp = |\Gamma_\perp|^2, \tag{8.77}$$

类似地，对于平行偏振

$$R_\parallel = \frac{P_\parallel^{\mathrm{r}}}{P_\parallel^{\mathrm{i}}} = |\Gamma_\parallel|^2. \tag{8.78}$$

透射率 T（或光学中的透射率）定义为透射功率与入射功率之比：

$$T_\perp = \frac{P_\perp^{\mathrm{t}}}{P_\perp^{\mathrm{i}}} = \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{|E_{\perp 0}^{\mathrm{i}}|^2}\frac{\eta_1}{\eta_2}\frac{A\cos\theta_{\mathrm{t}}}{A\cos\theta_{\mathrm{i}}} = |\tau_\perp|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right), \tag{8.79a}$$

$$T_\parallel = \frac{P_\parallel^{\mathrm{t}}}{P_\parallel^{\mathrm{i}}} = |\tau_\parallel|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right). \tag{8.79b}$$

► 入射波、反射波和透射波不必遵守电场守恒定律、磁场守恒定律或功率密度守恒定律，但它们必须遵守功率守恒定律。 ◄

事实上，在许多情况下，传输电场大于入射电场。功率守恒要求入射功率等于反射功率和传输功率之和。也就是说，对于垂直偏振，

$$P_\perp^{\mathrm{i}} = P_\perp^{\mathrm{r}} + P_\perp^{\mathrm{t}}, \tag{8.80}$$

或者

$$\frac{|E_{\perp 0}^{\mathrm{i}}|^2}{2\eta_1} A\cos\theta_{\mathrm{i}} = \frac{|E_{\perp 0}^{\mathrm{r}}|^2}{2\eta_1} A\cos\theta_{\mathrm{r}} + \frac{|E_{\perp 0}^{\mathrm{t}}|^2}{2\eta_2} A\cos\theta_{\mathrm{t}}. \tag{8.81}$$

使用等式。 (8.76)、(8.79a) 和 (8.79b) 导致

$$R_\perp + T_\perp = 1, \tag{8.82a}$$

$$R_\parallel + T_\parallel = 1, \tag{8.82b}$$

或者

$$|\Gamma_\perp|^2 + |\tau_\perp|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right) = 1, \tag{8.83a}$$

$$|\Gamma_\parallel|^2 + |\tau_\parallel|^2\left(\frac{\eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}}}\right) = 1. \tag{8.83b}$$

图 8-19 显示了空气-玻璃界面中 $(R_\parallel, T_\parallel)$ 与 θ_{i} 的函数关系图。请注意，$R_\parallel$ 和 $T_\parallel$ 之和始终等于 1，如方程式 1 所要求的那样。 (8.82b)。我们还注意到，在布儒斯特角 θ_{B} 处，$R_\parallel = 0$ 且 $T_\parallel = 1$。

表 8-2 总结了正入射和斜入射的 Γ、τ、R 和 T 的通用表达式。

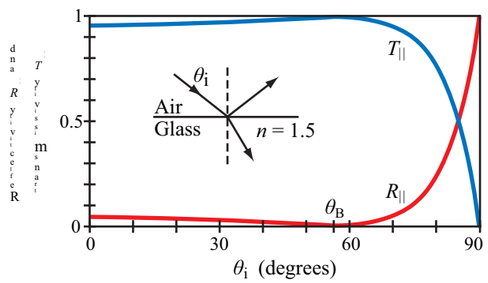


图 8-19 空气-玻璃界面的 $(R_\parallel, T_\parallel)$ 角度图。

Table 8-2 **Expressions for Γ, τ, *R*, and *T* for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2. Angles θ_i and θ_t are the angles of incidence and transmission, respectively.**

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \dfrac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_\perp = \dfrac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\Gamma_\parallel = \dfrac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Transmission coefficient	$\tau = \dfrac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_\perp = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\tau_\parallel = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\dfrac{\cos\theta_i}{\cos\theta_t}$
Reflectivity	$R = \lvert\Gamma\rvert^2$	$R_\perp = \lvert\Gamma_\perp\rvert^2$	$R_\parallel = \lvert\Gamma_\parallel\rvert^2$
Transmissivity	$T = \lvert\tau\rvert^2 \left(\dfrac{\eta_1}{\eta_2}\right)$	$T_\perp = \lvert\tau_\perp\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$T_\parallel = \lvert\tau_\parallel\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
Relation of *R* to *T*	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: (1) $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; (2) $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$; (4) for nonmagnetic media, $\eta_2/\eta_1 = n_1/n_2$.

Module 8.3 Oblique Incidence Upon specifying the frequency, polarization and incidence angle of a plane wave incident upon a planar boundary between two lossless media, this module displays vector information and plots of the reflection and transmission coefficients as a function of the incidence angle.

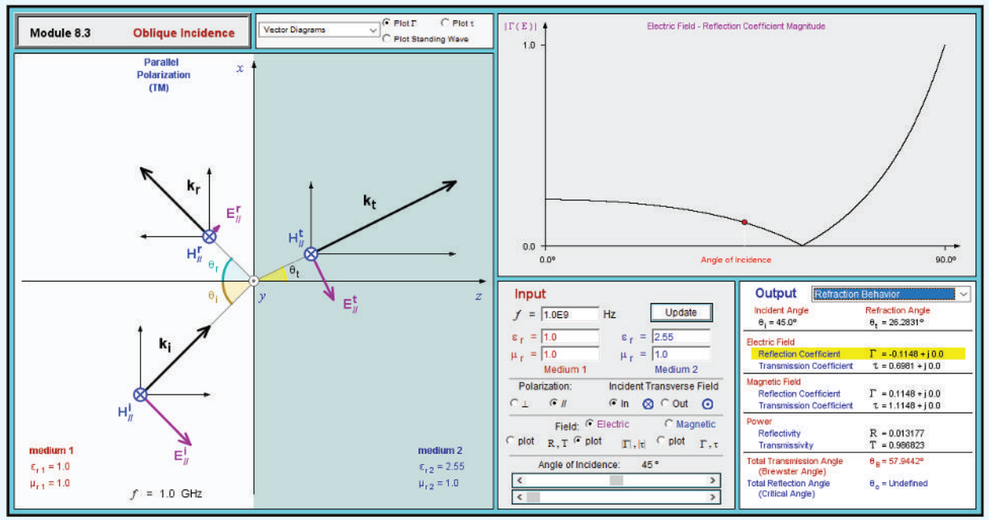
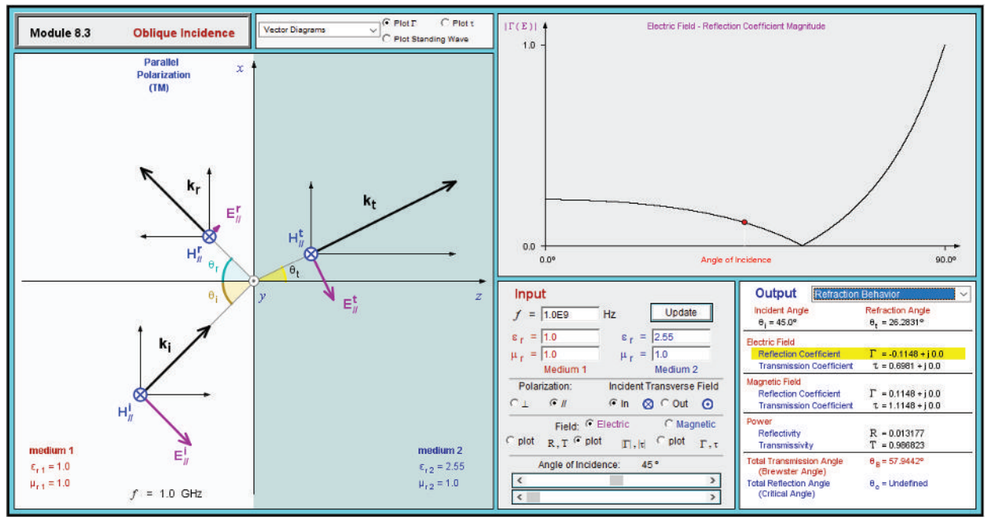


表 8-2 波从具有固有阻抗 η_1 的介质入射到具有固有阻抗 η_2 的介质时的 Γ、τ、R 和 T 表达式。角度θ_i和θ_t分别是入射角和透射角。

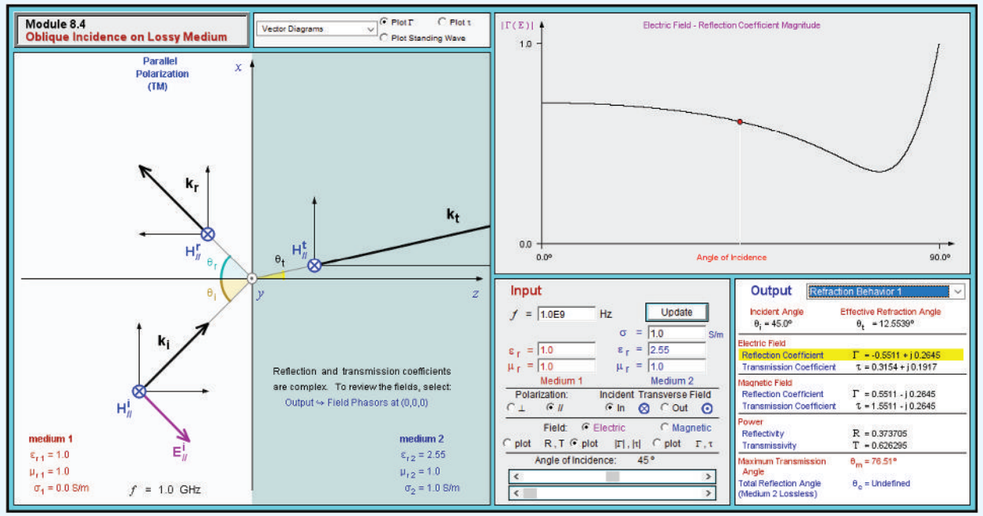
Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \dfrac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_\perp = \dfrac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\Gamma_\parallel = \dfrac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Transmission coefficient	$\tau = \dfrac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_\perp = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\tau_\parallel = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\dfrac{\cos\theta_i}{\cos\theta_t}$
Reflectivity	$R = \lvert\Gamma\rvert^2$	$R_\perp = \lvert\Gamma_\perp\rvert^2$	$R_\parallel = \lvert\Gamma_\parallel\rvert^2$
Transmissivity	$T = \lvert\tau\rvert^2 \left(\dfrac{\eta_1}{\eta_2}\right)$	$T_\perp = \lvert\tau_\perp\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$T_\parallel = \lvert\tau_\parallel\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
Relation of *R* to *T*	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: (1) $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; (2) $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$; (4) for nonmagnetic media, $\eta_2/\eta_1 = n_1/n_2$.

模块 8.3 倾斜入射 在指定入射到两个无损介质之间的平面边界上的平面波的频率、偏振和入射角后，该模块显示矢量信息以及作为入射角函数的反射和透射系数图。



Module 8.4 Oblique Incidence in Lossy Medium This module extends the capabilities of Module 8.1 to situations in which medium 2 is lossy.



Example 8-7: Beam of Light

A 5 W beam of light with circular cross section is incident in air upon the plane boundary of a dielectric medium with index of refraction of 5. If the angle of incidence is 60° and the incident wave is parallel polarized, determine the transmission angle and the powers contained in the reflected and transmitted beams.

Solution: From Eq. (8.56),

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i = \frac{1}{5}\sin 60^\circ = 0.17$$

or

$$\theta_t = 10^\circ.$$

With $\varepsilon_2/\varepsilon_1 = n_2^2/n_1^2 = (5)^2 = 25$, the reflection coefficient for parallel polarization follows from Eq. (8.68) as

$$\Gamma_\parallel = \frac{-(\varepsilon_2/\varepsilon_1)\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}{(\varepsilon_2/\varepsilon_1)\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}$$

$$= \frac{-25\cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}}{25\cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}} = -0.435.$$

The reflected and transmitted powers therefore are

$$P_\parallel^r = P_\parallel^i|\Gamma_\parallel|^2 = 5(0.435)^2 = 0.95 \text{ W},$$

$$P_\parallel^t = P_\parallel^i - P_\parallel^r = 5 - 0.95 = 4.05 \text{ W}.$$

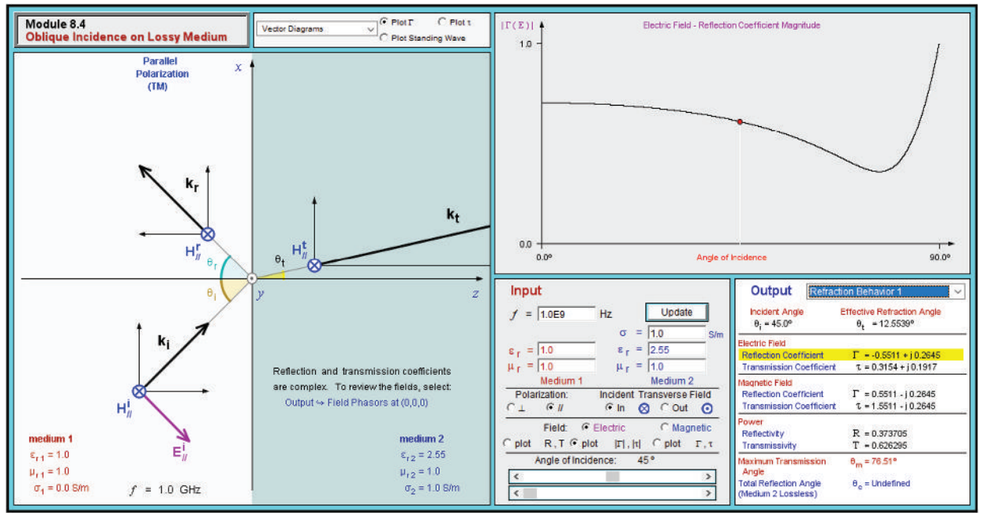
8-6 Waveguides

Earlier in Chapter 2, we considered two families of transmission lines, namely those that support ***transverse-electromagnetic*** (TEM) modes, and those that do not. Transmission lines belonging to the TEM family (**Fig. 2-4**), including coaxial, two-wire, and parallel-plate lines, support **E** and **H** fields that are orthogonal to the direction of propagation. Fields supported by lines in the other group—often called ***higher-order transmission lines***—may have **E** *or* **H** orthogonal to the direction of propagation $\hat{\mathbf{k}}$, but not both simultaneously. Thus, *at least one component of* **E** *or* **H** *is along* $\hat{\mathbf{k}}$.

► If **E** is transverse to $\hat{\mathbf{k}}$ but **H** is not, we call it a ***transverse electric*** (TE) mode, and if **H** is transverse to $\hat{\mathbf{k}}$ but **E** is not, we call it a ***transverse magnetic*** (TM) mode. ◄

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模块 8.4 有损介质中的斜入射 该模块将模块 8.1 的功能扩展到介质 2 有损的情况。



例8-7：光束

具有圆形横截面的 5 W 光束在空气中入射到折射率为 5 的介电介质的平面边界上。如果入射角为 60° 并且入射波为平行偏振，请确定透射角以及反射和透射光束中包含的功率。

解决方案：从方程。 (8.56)，

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i = \frac{1}{5}\sin 60^\circ = 0.17$$

或者

$$\theta_t = 10^\circ.$$

当 $\varepsilon_2/\varepsilon_1 = n_2^2/n_1^2 = (5)^2 = 25$ 时，平行偏振的反射系数可从等式得出： (8.68) 作为

$$\Gamma_\parallel = \frac{-(\varepsilon_2/\varepsilon_1)\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}{(\varepsilon_2/\varepsilon_1)\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}$$

$$= \frac{-25\cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}}{25\cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}} = -0.435.$$

因此，反射功率和传输功率为

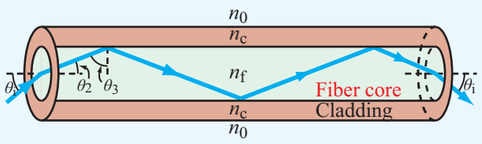
$$P_\parallel^r = P_\parallel^i|\Gamma_\parallel|^2 = 5(0.435)^2 = 0.95 \text{ W},$$

$$P_\parallel^t = P_\parallel^i - P_\parallel^r = 5 - 0.95 = 4.05 \text{ W}.$$

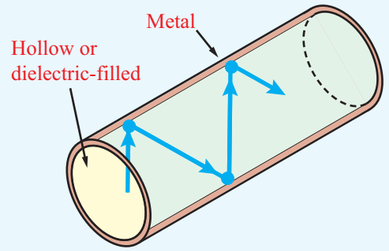
8-6 波导

在第 2 章前面，我们考虑了两类传输线，即支持横向电磁 (TEM) 模式的传输线和不支持横向电磁模式的传输线。属于 TEM 系列的传输线（图 2-4），包括同轴、两线和平行板线，支持与传播方向正交的 E 和 H 场。另一组中的线（通常称为高阶传输线）支持的场可能具有与传播方向 $\hat{k}$ 正交的 E 或 H，但不能同时两者。因此，E或H的至少一个分量沿着$\hat{k}$。

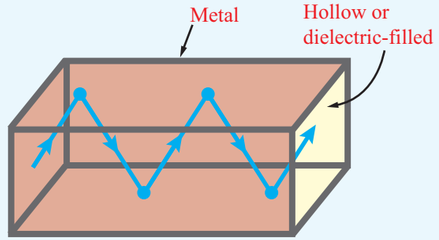
► 如果 E 横向于 $\hat{k}$ 但 H 不横向，我们将其称为横向电 (TE) 模式，如果 H 横向于 $\hat{k}$ 但 E 不横向，我们将其称为横向磁 (TM) 模式。 ◄



(a) Optical fiber

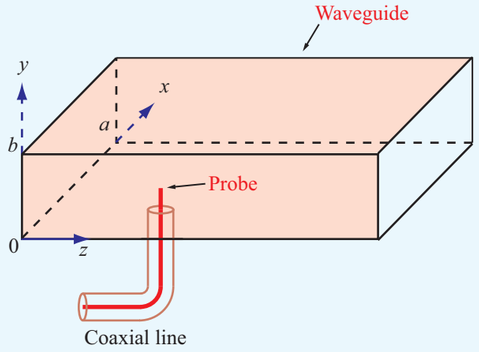


(b) Circular waveguide

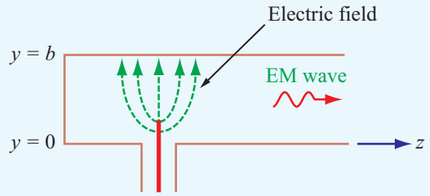


(c) Rectangular waveguide

Figure 8-20 Wave travel by successive reflections in (a) an optical fiber, (b) a circular metal waveguide, and (c) a rectangular metal waveguide.



(a) Coax-to-waveguide coupler

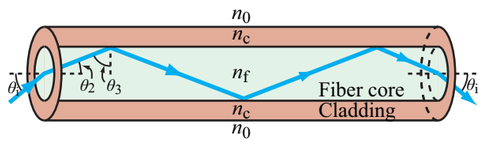


(b) Cross-sectional view at $x = a/2$

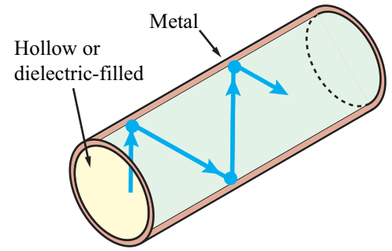
Figure 8-21 The inner conductor of a coaxial cable can excite an EM wave in the waveguide.

Among all higher-order transmission lines, the two most commonly used are optical fiber and the metal waveguide. As noted in Section 8-3, a wave is guided along an optical fiber through successive zigzags by taking advantage of the total internal reflection at the boundary between the (inner) core and the (outer) cladding (**Fig. 8-20(a)**). Another way to achieve internal reflection at the core's boundary is to have its surface coated by a conducting material. Under the proper conditions (we shall elaborate later) a wave excited in the interior of a hollow conducting pipe, such as the circular or rectangular waveguides shown in **Figs. 8-20(b)** and **(c)**, undergoes a process similar to that of successive internal reflection in an optical fiber, resulting in propagation down the pipe. Most waveguide applications call for air-filled guides, but in some cases, the waveguide may be filled with a dielectric material used to alter its propagation velocity or impedance, or it may be vacuum-pumped to eliminate air molecules in order to prevent voltage breakdown, thereby increasing its power-handling capabilities.

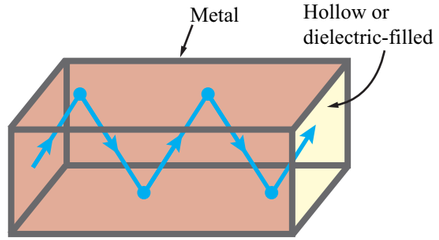
Figure 8-21 illustrates how a coaxial cable can be connected to a rectangular waveguide. With its outer conductor connected to the metallic waveguide enclosure, the coaxial cable's inner conductor protrudes through a tiny hole into the waveguide's interior (without touching the conducting surface). Time-varying electric field lines extending between the protruding inner conductor and the inside surface of the guide provide the excitation necessary to transfer a signal from the coaxial



(a) Optical fiber

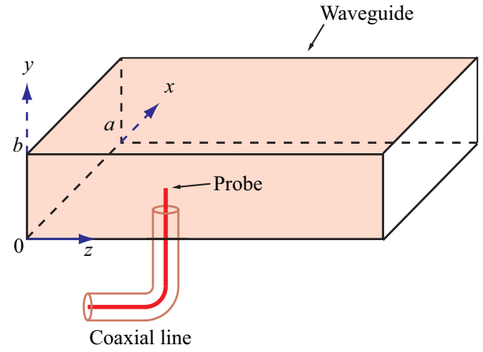


(b) Circular waveguide

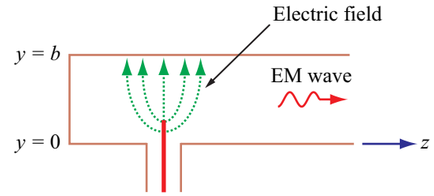


(c) Rectangular waveguide

图 8-20 波在 (a) 光纤、(b) 圆形金属波导和 (c) 矩形金属波导中通过连续反射传播。



(a) Coax-to-waveguide coupler



(b) Cross-sectional view at $x = a/2$

图8-21 同轴电缆的内导体可以在波导中激发电磁波。

在所有高阶传输线中，最常用的两种是光纤和金属波导。如第 8-3 节所述，利用（内）纤芯和（外）包层之间边界处的全内反射，波被引导通过连续的锯齿形光纤（图 8-20(a)）。在核心边界实现内部反射的另一种方法是在其表面涂上导电材料。在适当的条件下（我们稍后将详细说明），在

空心导管的内部，例如图1和图2所示的圆形或矩形波导。 8-20(b) 和 (c) 经历了类似于光纤中连续内部反射的过程，导致沿着管道传播。大多数波导应用需要充气波导，但在某些情况下，波导可能填充有介电材料，用于改变其传播速度或阻抗，或者可以对其进行真空泵浦以消除空气分子，以防止电压击穿，从而提高其功率处理能力。

图 8-21 说明了如何将同轴电缆连接到矩形波导。同轴电缆的外导体连接到金属波导外壳，内导体通过一个小孔伸入波导内部（不接触导电表面）。在突出的内导体和导管内表面之间延伸的时变电场线提供了从同轴电缆传输信号所需的激励。